

END-TO-END SUPPLY CHAIN PERFORMANCE ANALYTICS

Project Report

An end-to-end analysis of vendor profitability, procurement and logistics, sales trends, store performance, inventory integrity, and dead stock.

Prepared By

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End-to-End Data Analytics Project

1. Executive Summary

This project analyzes vendor, product, purchase, sales, store, and inventory data to understand where the business is making money, where cash is tied up, and where operational data or processes need attention.

The analysis covers 2024 and is built from SQL summary tables together with the raw sales and purchases data. The analysis notebook contains the detailed calculations and validation; this report presents the main business findings in a simpler format.

Key numbers

Metric	Result
Total purchases	\$321.90M
Total sales	\$451.72M
Gross profit	\$129.82M
Overall profit margin	28.74%
Top 10 vendor share of purchases	65.3%
Unsold inventory capital	\$15.60M
Company-wide dead stock	178 products / \$282.18K
Store-level products unsold at at least one store	2,556
Invoiced POs with no matching purchase record	1,336 of 5,543

Main takeaways

- A relatively small group of vendors accounts for a large share of purchases, sales, and gross profit. This makes vendor concentration important to monitor.
- The biggest inventory issue is broader unsold stock (\$15.60M), rather than company-wide dead stock alone. Store-level unsold products are also much more common than products that never sell anywhere.
- Classification 2 has a higher sales-weighted margin than Classification 1 (35.1% vs. 25.2%). The meaning of this classification is not documented in the source data and should be confirmed.
- Low-selling vendor-product lines are a major profitability concern: their sales-weighted margin is about -87.5%, compared with about 29.3% for the top-selling tier.
- Procurement data has one important reconciliation issue: 1,336 invoiced POs have no matching purchase/receiving record, representing \$90.31M of invoiced value.
- Sales peak in December and are lowest in February. Receiving activity is also strongly concentrated on particular weekdays.

2. Project Scope and Approach

The notebook is structured as an end-to-end business analysis rather than a single visualization exercise. It covers seven areas: vendor-product profitability, procurement and logistics, time-series trends, store and city performance, inventory reconciliation, dead stock, and final recommendations.

Data and analytical grain

- Vendor-product analysis uses one row per VendorNumber and ProductID from vendor_product_sales_summary.
- Procurement analysis uses purchase-order level data and examines freight, receiving, invoicing, and payment timing.
- Store and inventory analysis uses Store + ProductID level information where required.
- Inventory reconciliation compares beginning inventory, purchases, sales, and ending inventory.
- Dead stock is kept separately because products with zero sales are intentionally retained for this analysis.

Important analytical choices

- Segment profitability is compared using sales-weighted margins rather than simple averages, because a simple average can give tiny products the same importance as major products.
- Freight is analyzed at vendor/PO level because the vendor-level freight field is repeated across product rows.
- For the main profitable-product analysis, rows with non-positive gross profit, non-positive margin, or zero sales quantity are excluded. Dead-stock analysis then brings zero-sales products back in explicitly.
- Statistical testing is used where the notebook compares top-selling and low-selling margin distributions.

3. Vendor Profitability and Product Performance

3.1 Overall business performance

The business generated \$451.72M in sales from \$321.90M of purchases, producing \$129.82M gross profit and an overall sales-weighted margin of 28.74%.

3.2 Vendor concentration

The top 10 vendors account for 65.3% of total purchases. The same group generates about 65.0% of sales and 64.2% of gross profit. This means vendor concentration is significant across the business, not only on the purchasing side.

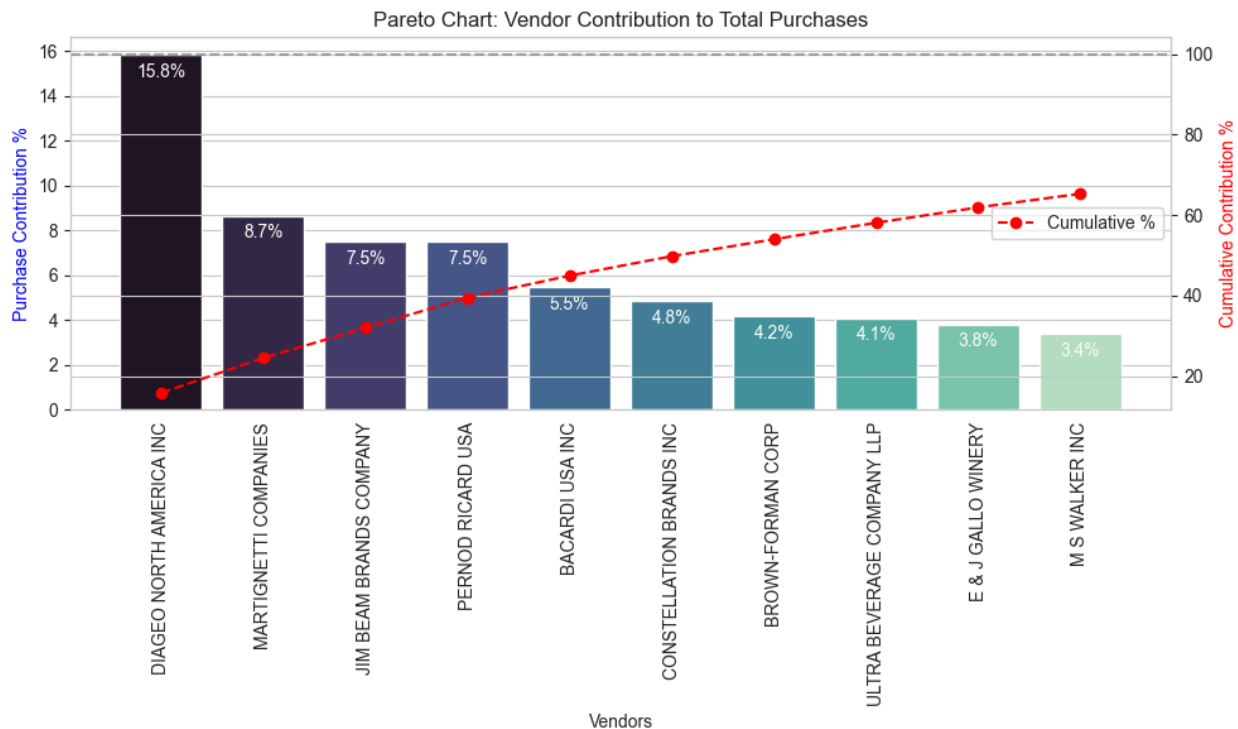


Figure 1. Top 10 vendors' share of total purchases

3.3 Vendor profitability

Among vendors with at least \$10K in sales, profitability varies widely. The highest-margin vendors include ALISA CARR BEVERAGES, THE PIERPONT GROUP LLC and R.P.IMPORTS INC. At the other end, ADAMBA IMPORTS INTL INC. has a negative margin of about -13.6%. These comparisons should be interpreted together with sales volume because a very high margin on a small sales base does not have the same business impact as a moderate margin on a large sales base.

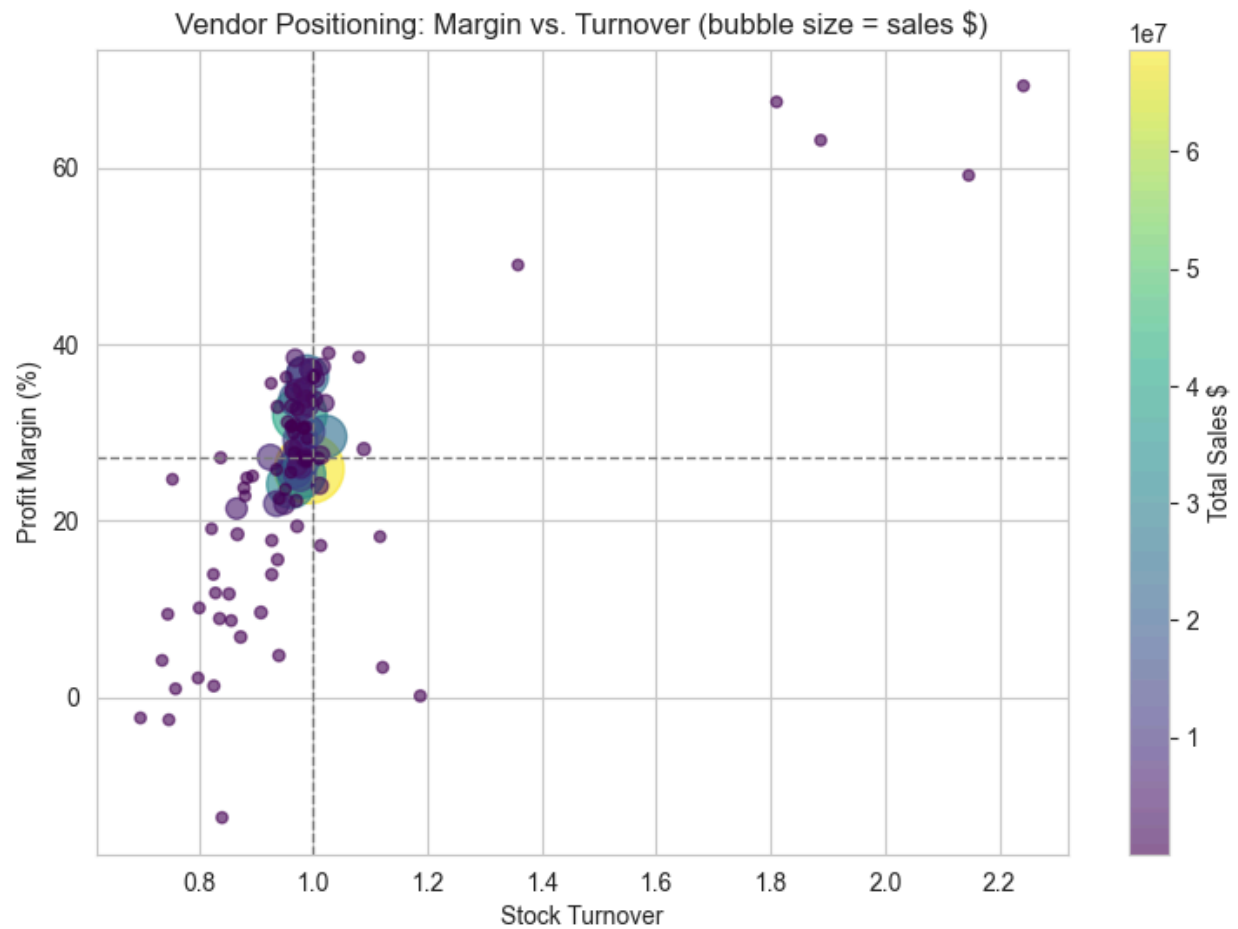


Figure 2. Vendor margin and stock turnover

Vendor margin and stock turnover have a correlation of 0.71 in the vendor-level analysis. This indicates that the two measures move together to some extent, although turnover alone does not explain profitability.

3.4 Low-selling products and profitability

The analysis identifies 105 products with very low sales and high margins that could be candidates for targeted pricing or promotion. More importantly, the top-selling and low-selling groups show a large profitability difference.

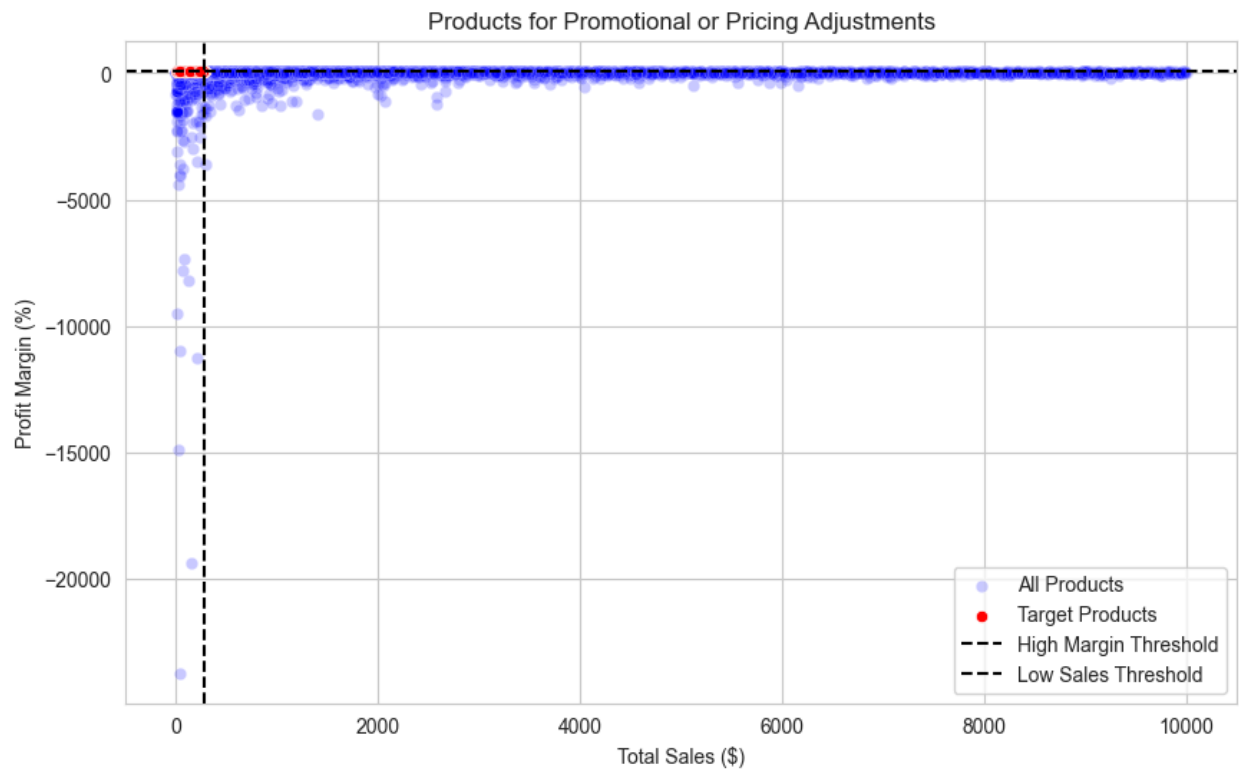


Figure 3. Products flagged for low sales and high margin

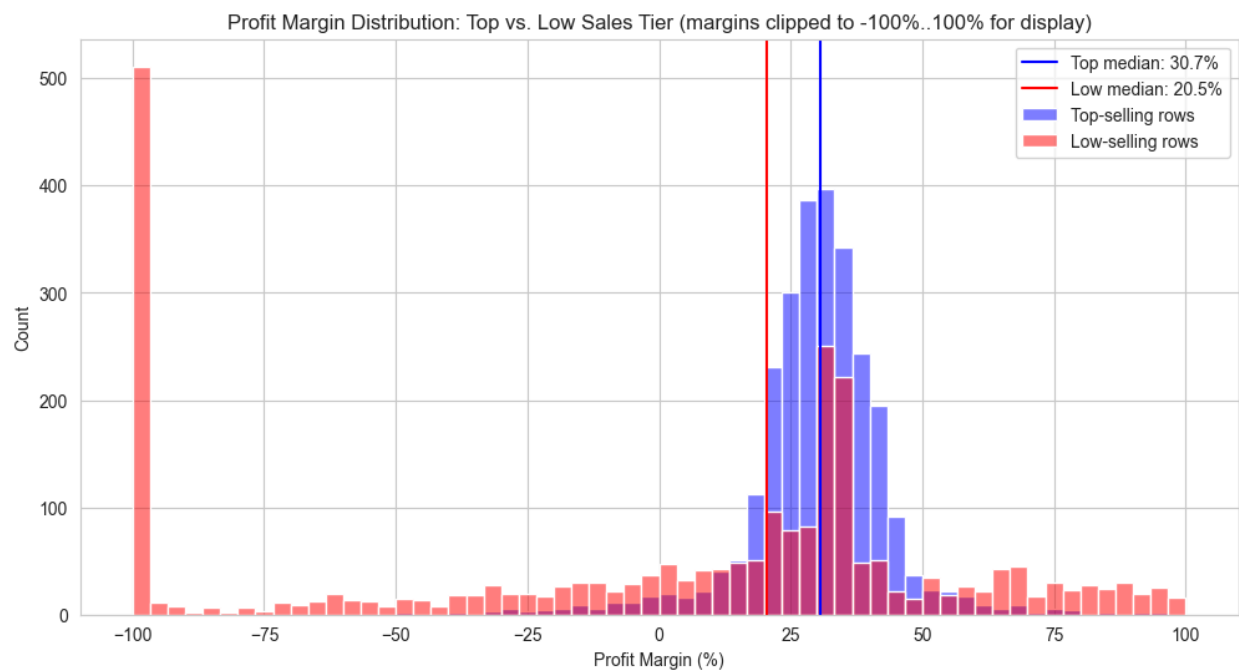


Figure 4. Margin comparison between top-selling and low-selling vendor-product rows

Top-selling rows have a sales-weighted margin of 29.25%, while low-selling rows have a sales-weighted margin of -87.52%. A Mann-Whitney U test gives a p-value of 3.033e-70, so the notebook rejects the null hypothesis that the two margin distributions are the same.

3.5 Product pricing

About 26.6% of products have an actual selling price that differs from the catalog price by more than \$1. Actual selling price also has almost no correlation with total sales (-0.015), so higher prices do not automatically translate into higher revenue.

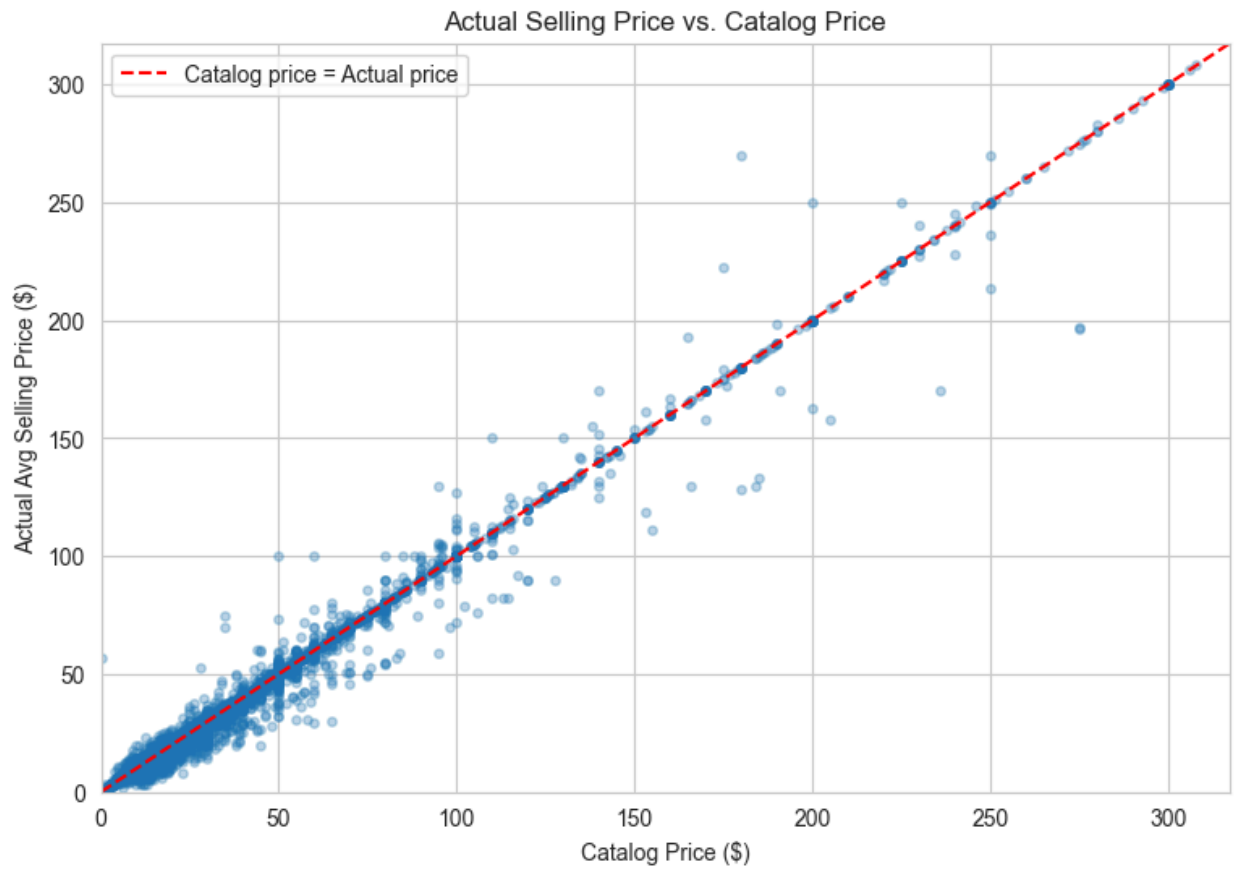


Figure 5. Actual selling price vs. catalog price analysis

4. Purchasing, Bulk Buying and Logistics

4.1 Bulk purchasing and unit cost

The average unit purchase price falls as order size increases: about \$43.78 for small orders, \$17.89 for medium orders, and \$11.31 for large orders. The pattern is consistent with a strong relationship between order size and lower observed unit cost. However, the notebook correctly notes that this does not by itself prove a pure volume-discount effect because product mix can also change with order size.

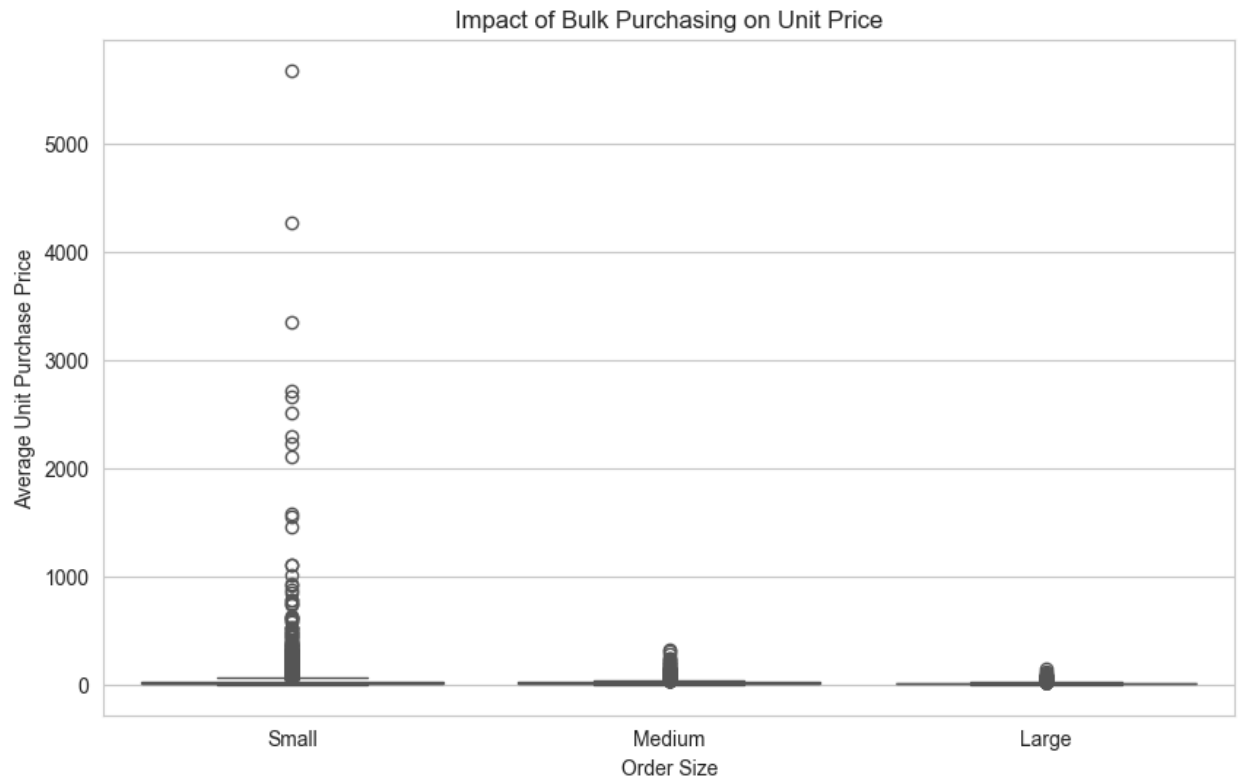


Figure 6. Unit purchase price by order size

4.2 Freight cost

Freight is a relatively small cost compared with total purchase value. The average freight is about 0.55% of PO value. Freight amount rises strongly with PO size (correlation 0.985), which is expected. Freight percentage is therefore the more useful measure when comparing vendors.

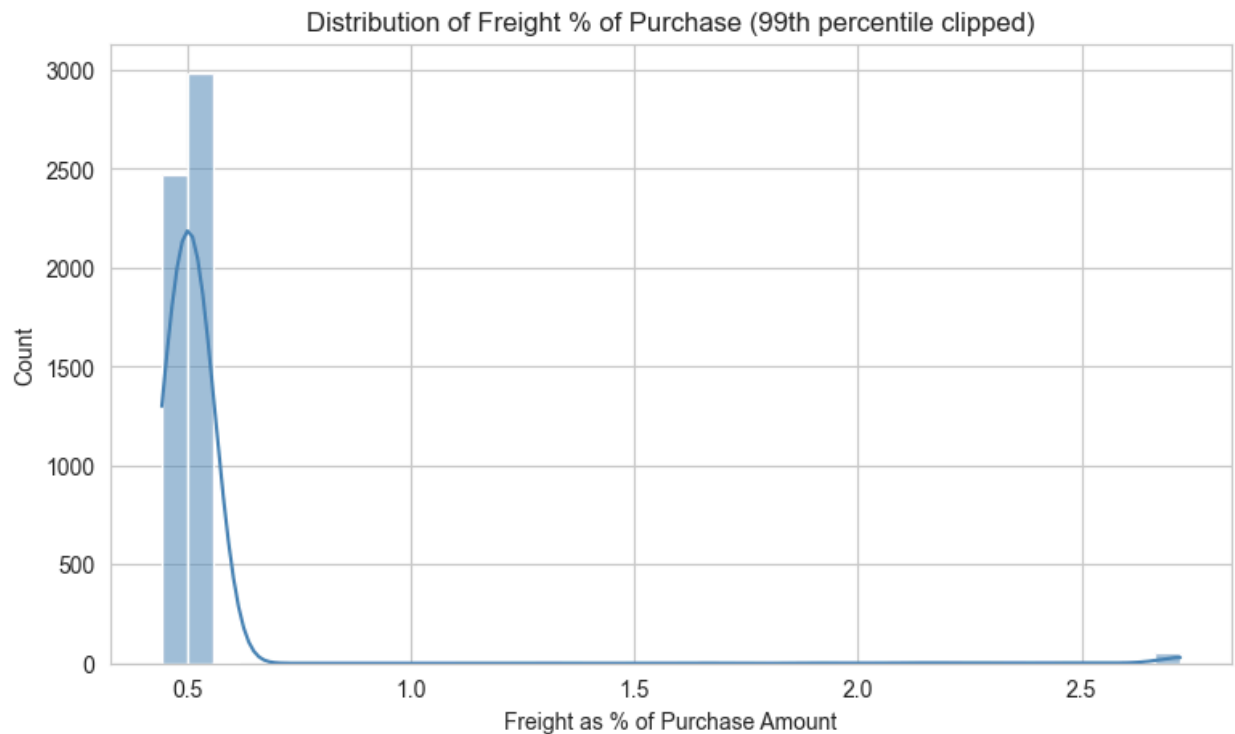


Figure 7. Freight as a percentage of purchase value

4.3 Lead times

For the 4,207 POs with receiving records, the median order-to-receive time is 6 days, receive-to-invoice is 7 days, and invoice-to-pay is 35 days. Average lead-time speed and consistency are different issues: their correlation is about 0.49.

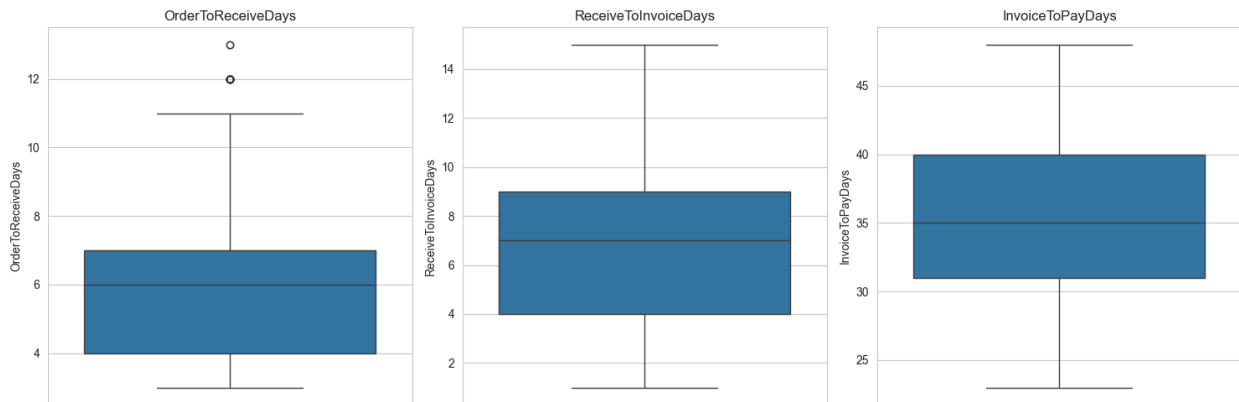


Figure 8. Procurement lead-time stages

4.4 Missing purchase/receiving records

This is one of the most important data-quality findings. 1,336 of 5,543 invoiced POs (24.1%) have no matching purchase record. Their combined invoiced amount is \$90.31M, or 28.1% of all invoiced dollars. The detailed check found zero cases where an existing purchase record had a different invoice quantity. In other words, the flagged cases are missing purchase/receiving records rather than quantity disagreements within matched records.

- This should be checked with the receiving and invoicing teams.

- Possible explanations include goods not yet received, missing extract data, or invoices recorded before receipts. The notebook does not select one explanation as proven.

5. Sales and Time-Series Trends

5.1 Sales seasonality

Sales show a clear seasonal pattern. December is the strongest month at \$52.31M, while February is the lowest at \$28.88M. This pattern can be considered when planning inventory, staffing, and purchasing.

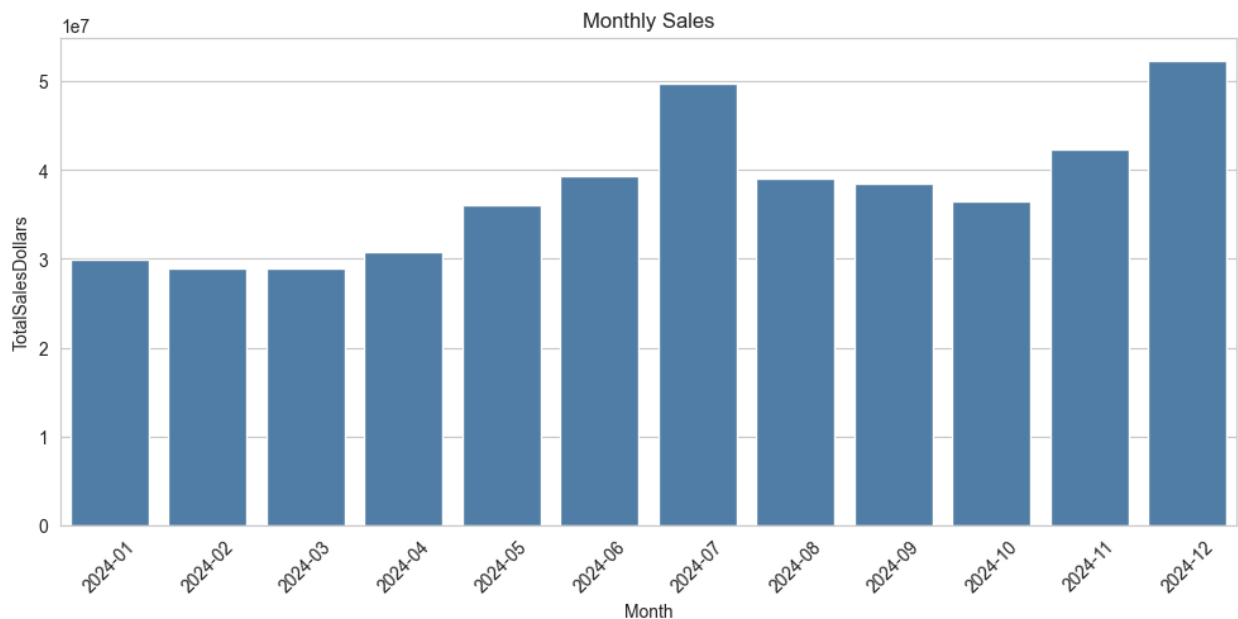


Figure 9. Monthly sales trend

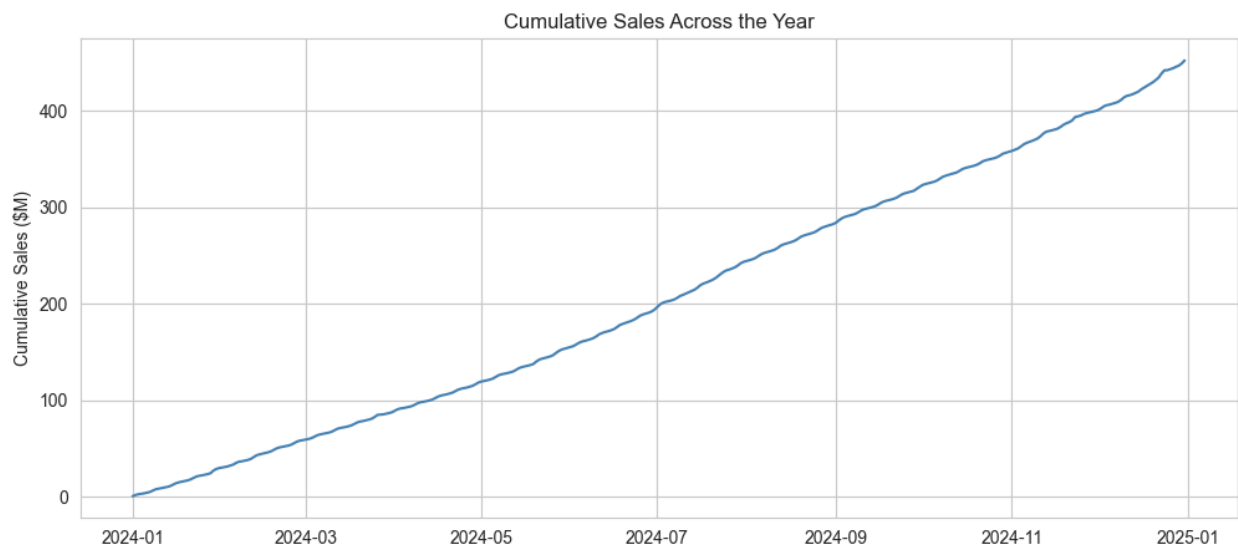


Figure 10. Cumulative sales trend

5.2 Weekly selling and receiving pattern

Receiving activity is strongly concentrated on certain weekdays. Very little stock is received on Tuesdays and Wednesdays, while other days generally average around \$0.9M to \$1.4M in receiving value. This looks like a delivery-scheduling pattern rather than random variation and should be compared with the actual logistics schedule.

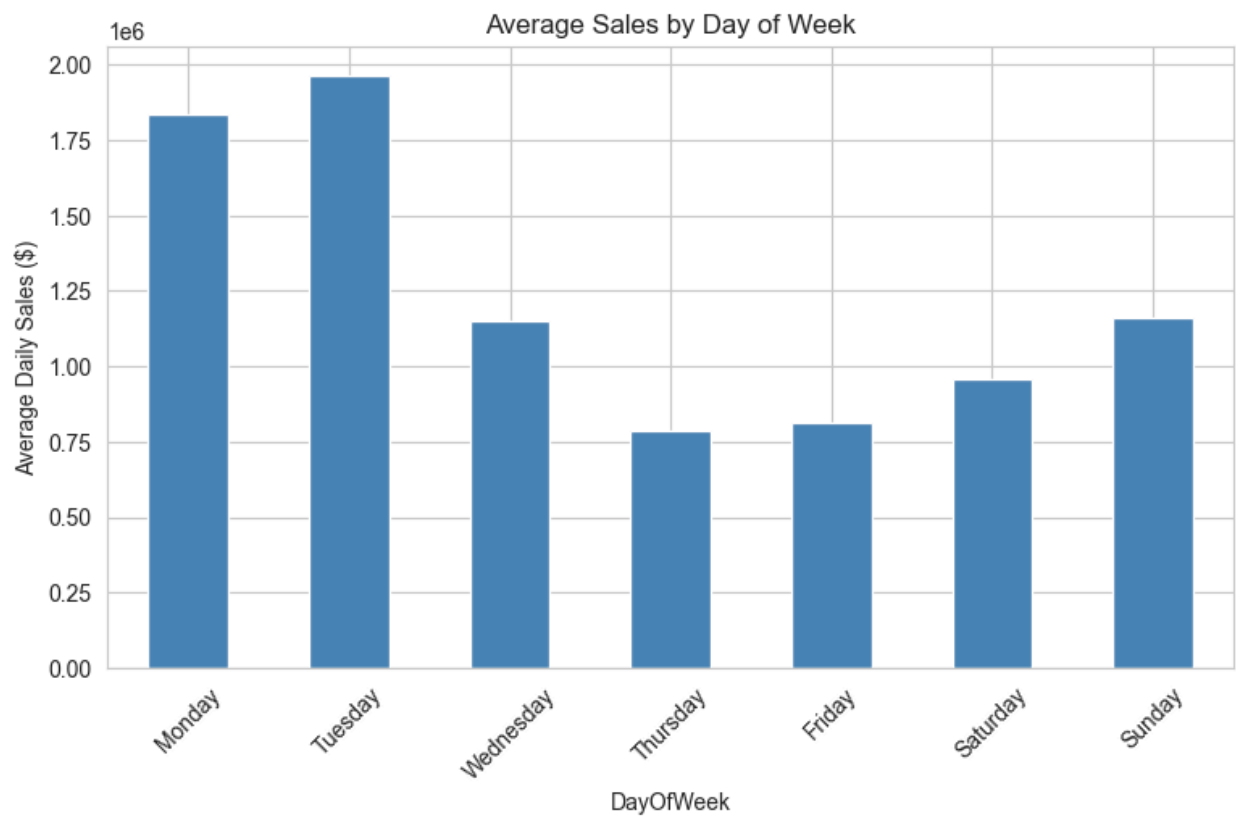


Figure 11. Sales pattern by day of week

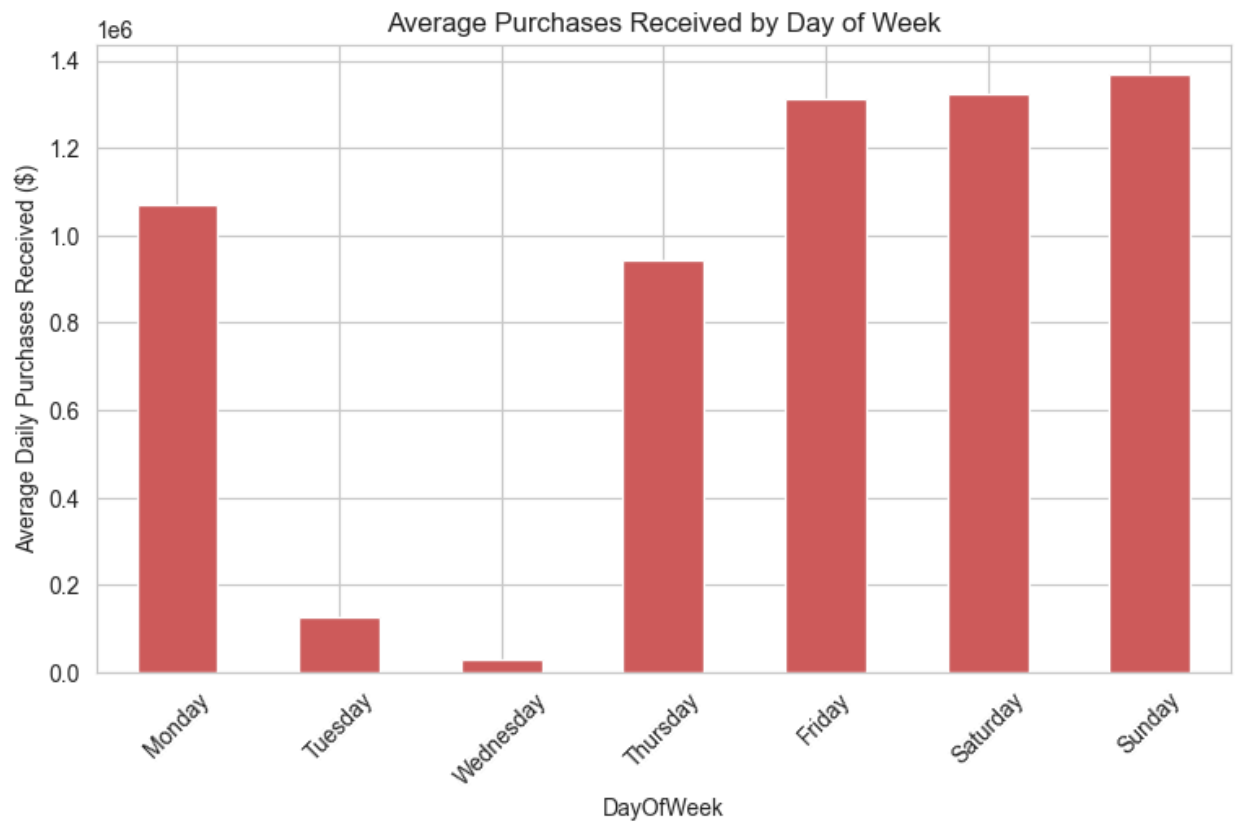


Figure 12. Purchases received by day of week

5.3 Promotions

Promotional/free sales are extremely rare in the data: only 55 transactions out of about 12.8 million sales rows, representing 103 units. There is therefore not enough promotional activity to measure its effect on sales reliably.

6. Store and City Performance

6.1 Store performance

Store performance varies considerably, while store size does not show a strong relationship with profit margin. This means stores should be reviewed individually rather than judged mainly by size.

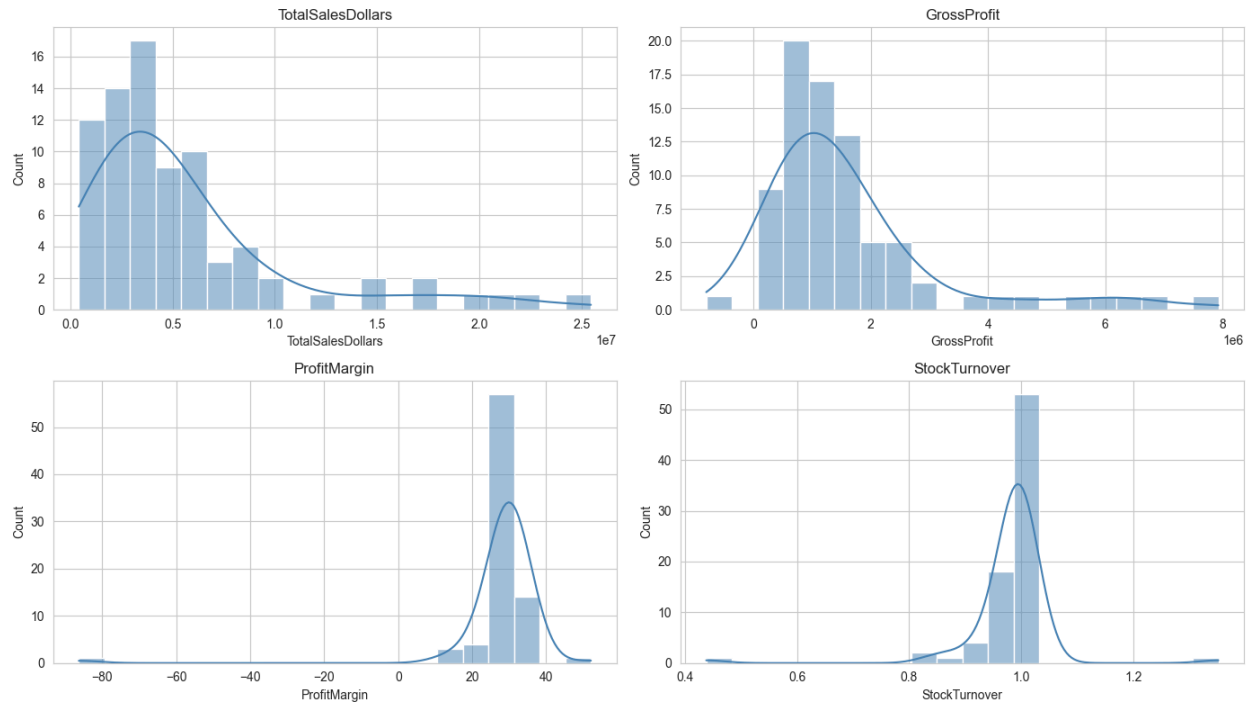


Figure 13. Store performance distribution

6.2 Store 81 and new/expanding stores

Store 81 has a profit margin of about -86%, far below the other stores. It also has 3,589 new products and no discontinued products. The five stores with the most new products (50, 81, 74, 41 and 44) are all among the ten lowest-margin stores. The notebook interprets this as potentially consistent with new or expanding stores building opening stock before sales catch up; this should be confirmed against store opening records.

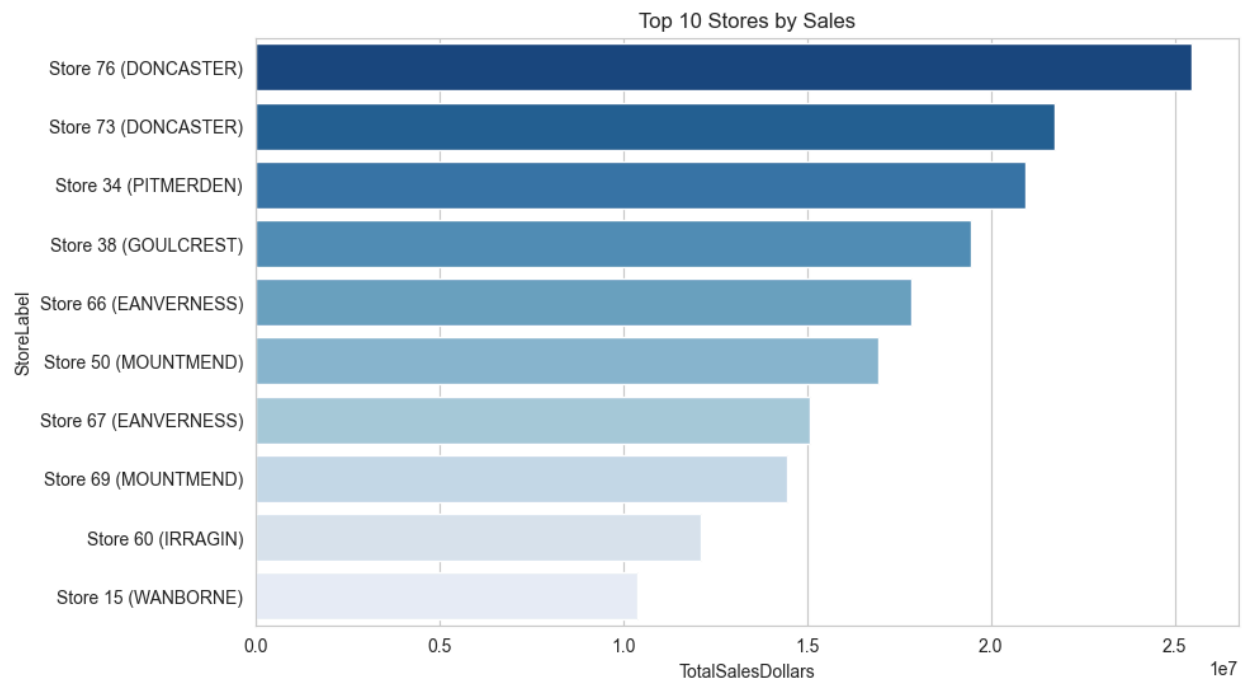


Figure 14. Top stores by sales

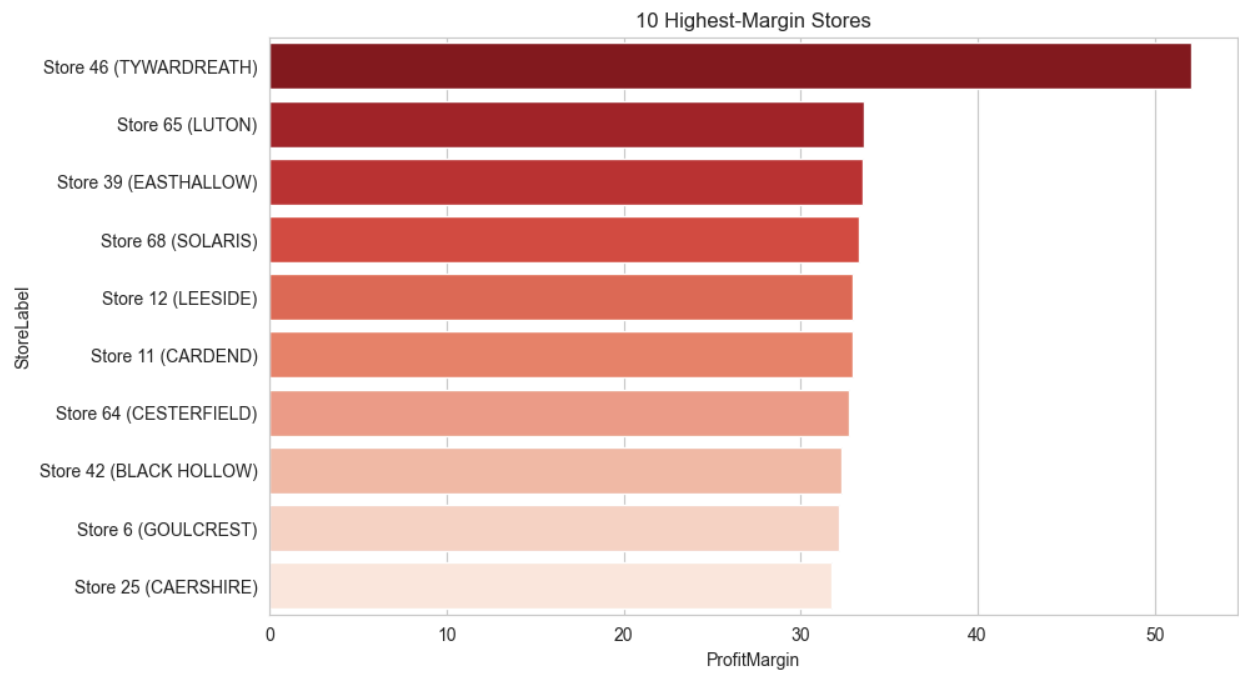


Figure 15. Bottom stores by sales

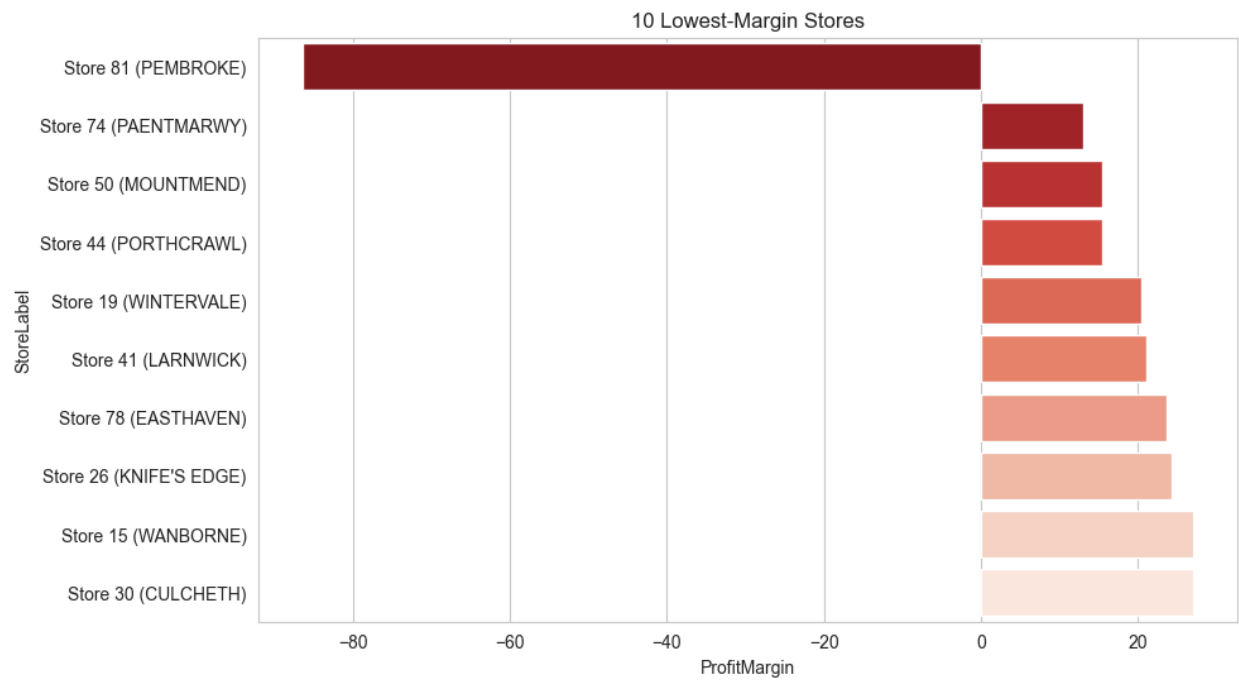


Figure 16. Store profit-margin comparison

6.3 City performance

City-level results vary, but cities contain different numbers and sizes of stores. City comparisons should therefore be considered together with store count and store-level performance.

7. Inventory Reconciliation and Stock Movement

7.1 Inventory grew during the year

Total inventory increased from 4,219,275 units at the start of the year to 4,885,776 units at the end of the year, a net increase of 666,501 units or 15.8%.

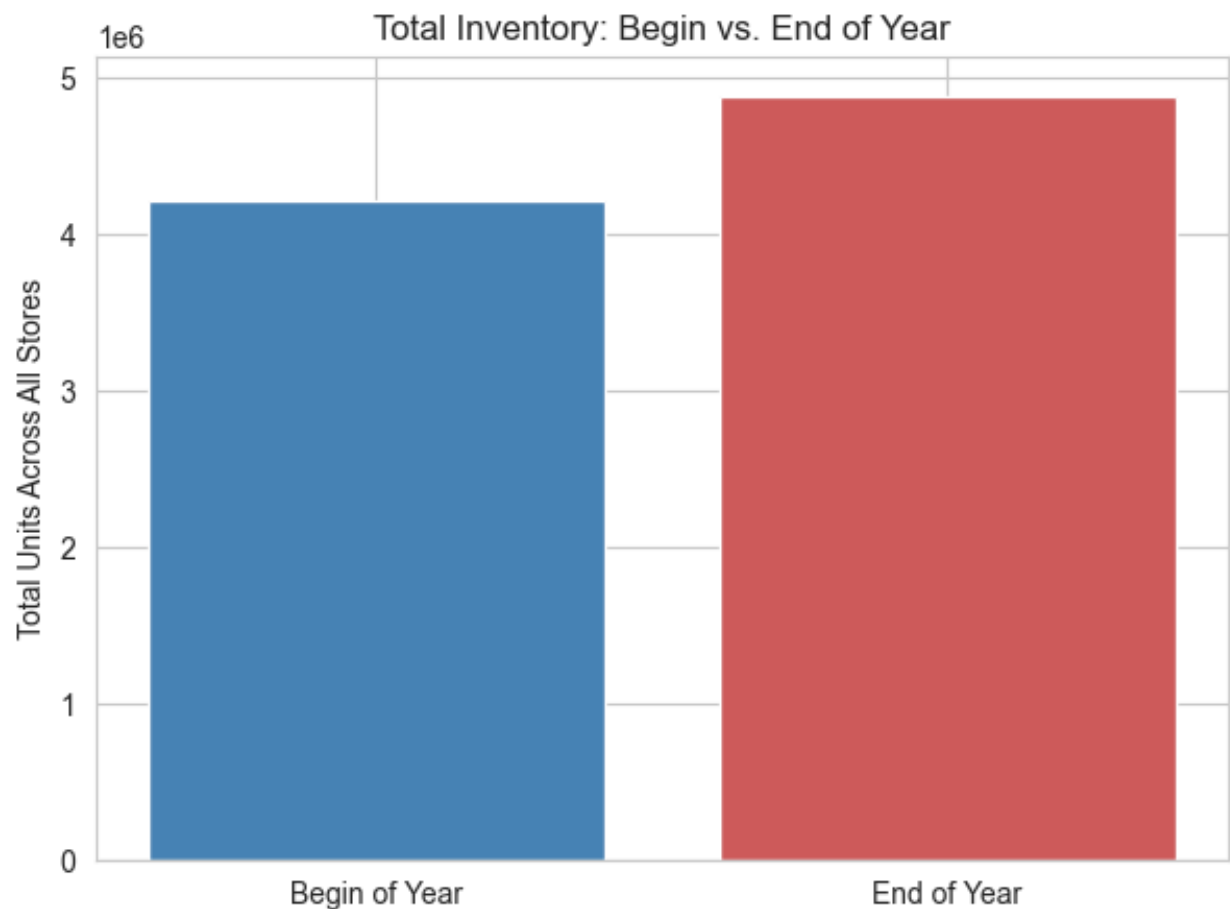


Figure 17. Beginning vs. ending inventory

7.2 Inventory reconciliation is very clean

99.9902% of Store-Product records reconcile exactly between beginning inventory, purchases, sales and ending inventory. Only 25 of 254,835 records have a non-zero variance, with a total variance of 207 units and a largest single difference of 36 units.

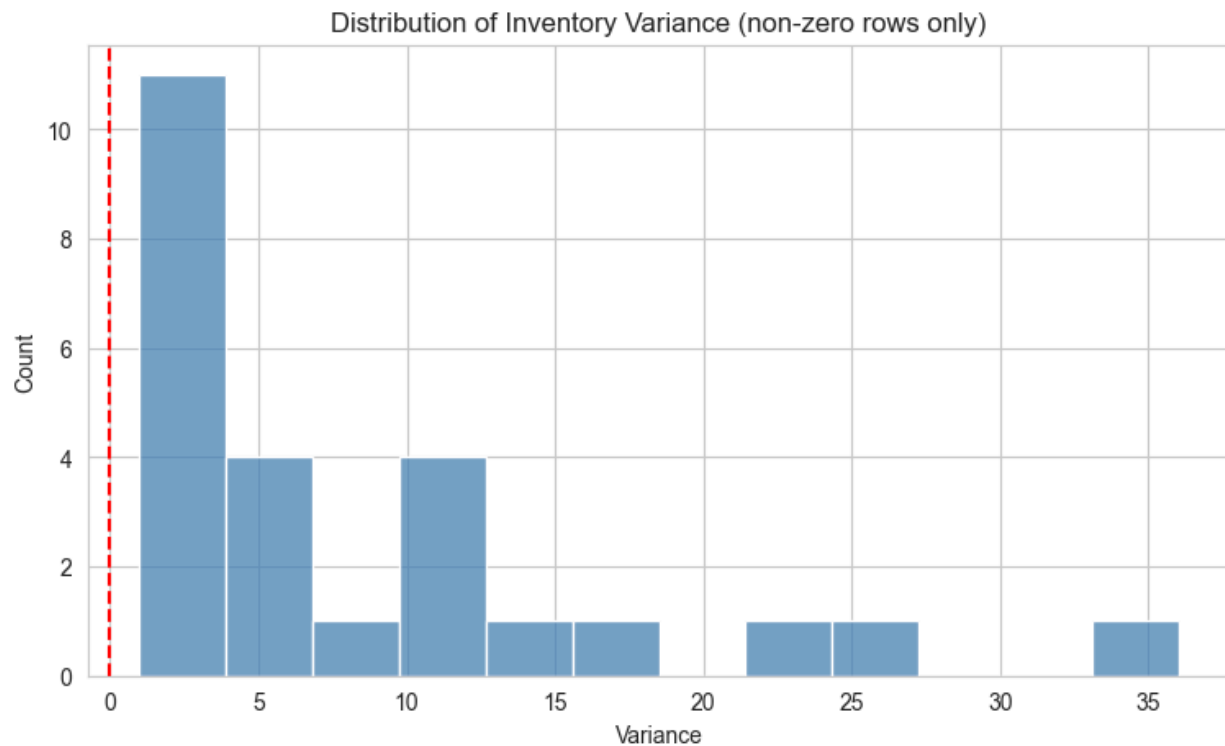


Figure 18. Inventory reconciliation variances

All 25 variances are positive. Several of the largest cases have negative expected ending stock, which suggests missing receipts or stock movements in the purchase data rather than straightforward shrinkage.

7.3 Product additions and discontinuations

There are 48,306 products that appear only in ending inventory and 30,346 that appear only in beginning inventory. Product churn differs substantially across vendors and stores. Store 81 again stands out with 3,589 new products and zero discontinued products.

7.4 Price movement

Among 176,183 Store-Product combinations with both beginning and ending prices, 29.6% had a price change. Changes ranged from a \$40 decrease to a \$260 increase.



Figure 19. Distribution of Store-Product price changes

8. Unsold Inventory and Dead Stock

8.1 General unsold inventory

The broader unsold inventory measure is \$15.60M. The largest amounts are associated with major vendors such as MARTIGNETTI COMPANIES (\$1.93M), DIAGEO NORTH AMERICA INC (\$1.66M), ULTRA BEVERAGE COMPANY LLP (\$1.48M), and JIM BEAM BRANDS COMPANY (\$1.14M). This is a larger business issue than company-wide dead stock alone.

8.2 Company-wide dead stock

Company-wide dead stock means products purchased but never sold anywhere during the year. There are 178 such products out of 10,693 vendor-product lines (1.7%), with about \$282.18K of capital tied up.

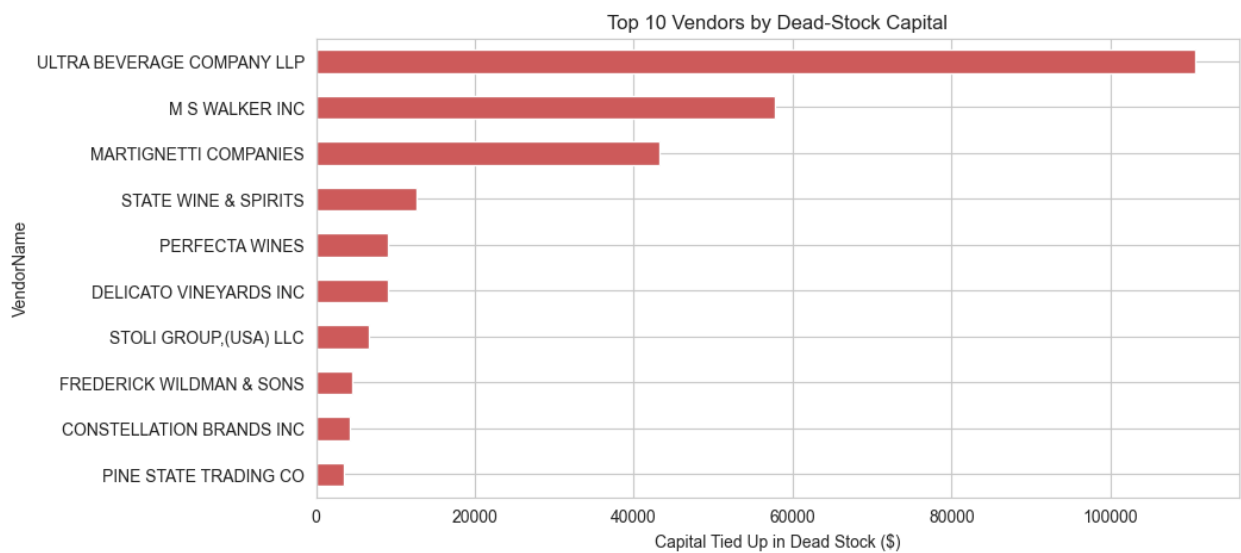


Figure 20. Vendors with the largest dead-stock dollar values

8.3 Dead-stock rate vs. dead-stock dollars

The vendors with the highest dead-stock rate are not the same as those with the highest dead-stock dollar value. Both measures are useful: dollar value identifies financial exposure, while rate identifies vendors where a larger share of the catalog is not moving.

8.4 Product type and dead stock

Classification 2 has a dead-stock rate of about 2.1% of its vendor-product lines compared with about 0.7% for Classification 1. Bottle-size differences are small and are based on relatively few dead-stock products, so they should not be over-interpreted.

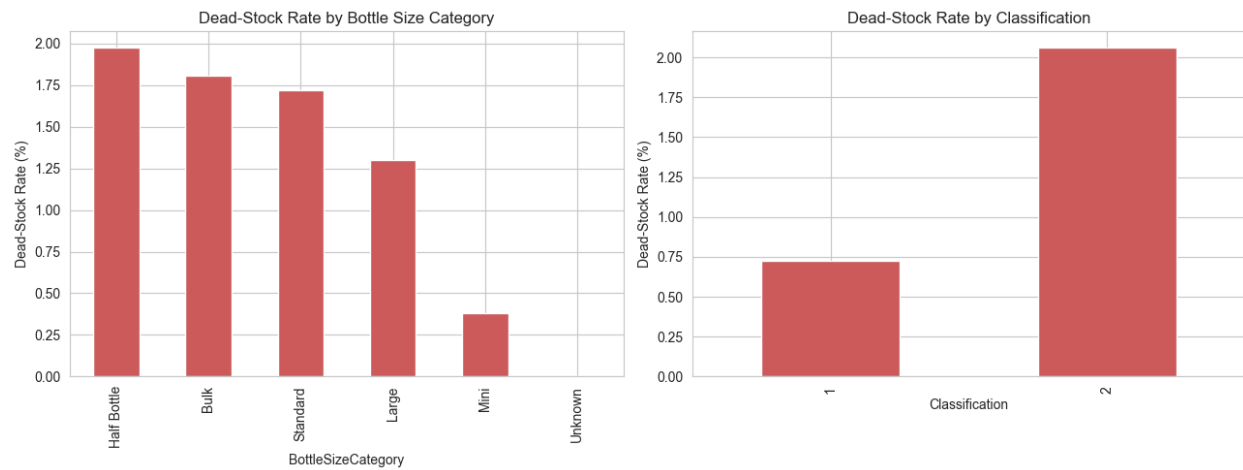


Figure 21. Dead-stock rates by classification and bottle size

8.5 Store-level dead stock

A much larger issue appears when dead stock is checked at individual stores. 2,556 products are unsold at least at one store, compared with only 178 products that are unsold across the whole company. Many of these products may still sell well at other stores, so store-to-store transfers or changes in local product mix could be useful.

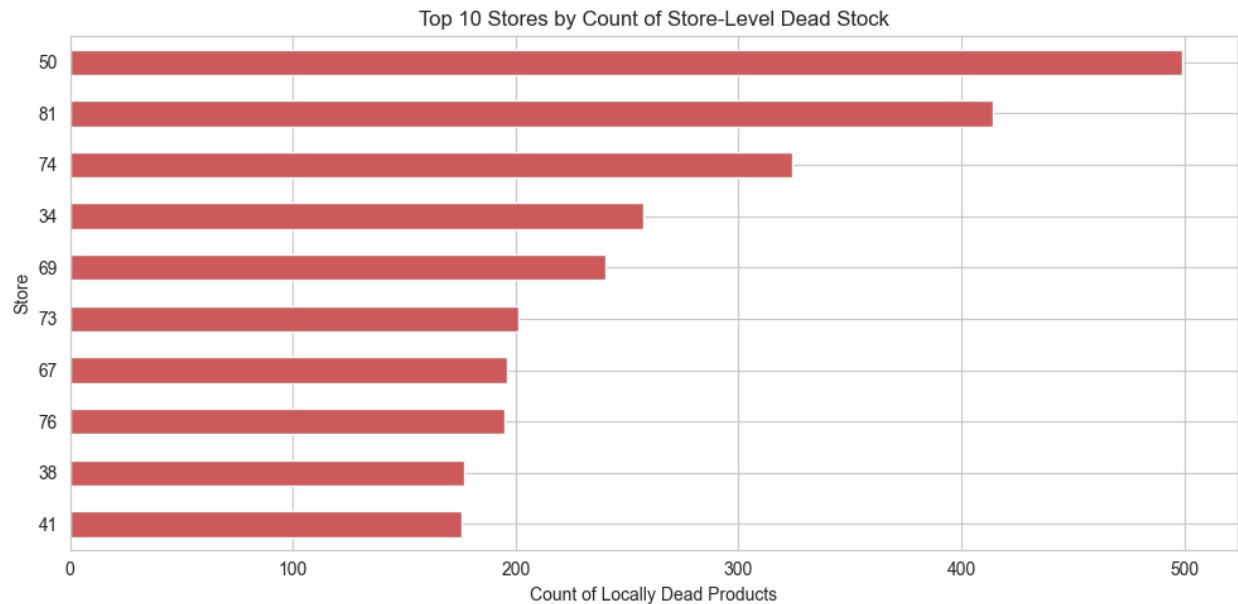


Figure 22. Store-level products with no local sales

9. Recommendations

- Monitor vendor concentration because the top 10 vendors account for 65.3% of purchases and a similar share of sales and gross profit.
- Prioritize low-selling vendor-product lines for pricing, assortment, or promotional review. Their aggregate margin is much weaker than the top-selling group.
- Confirm what Classification 1 and Classification 2 represent before using the large margin difference as a business decision driver.
- Investigate the 1,336 invoiced POs without matching purchase/receiving records and reconcile the \$90.31M of invoiced value.
- Use lead-time consistency, not only average lead time, when reviewing supplier reliability.
- Treat freight percentage as a secondary cost lever because freight is only about 0.55% of PO value on average.
- Plan inventory and staffing around the observed December sales peak and February sales low, while checking whether the weekday receiving pattern reflects the actual delivery calendar.
- Review the lowest-margin stores individually and confirm whether high new-product counts are related to store openings or expansion.
- Focus inventory actions on the \$15.60M of broader unsold inventory and on products that are not moving at specific stores.
- Use both dead-stock dollars and dead-stock rate when reviewing vendors and products.
- Review the 25 inventory reconciliation exceptions even though the overall reconciliation rate is extremely high.

10. Final Conclusion

Overall, the business is profitable at a 28.74% sales-weighted margin, but the analysis shows several areas where better inventory, vendor, and data controls could improve performance. The biggest opportunities are concentrated in unsold inventory, low-selling product lines, vendor concentration, store-level product allocation, and the missing purchase records behind a significant group of invoiced POs.

The analysis also shows that many operational patterns are measurable and actionable: sales seasonality, delivery-day concentration, vendor lead-time consistency, price changes, and product churn. The next step would be to validate the flagged cases with operational teams and use the findings to prioritize specific vendor, store, and inventory actions.

Appendix: Important Data Notes

- The notebook uses sales-weighted margins for segment comparisons because simple row averages can be misleading.
- Classification is not documented in the source data, so its business meaning should be confirmed before drawing operational conclusions from it.
- Dead stock and broader unsold inventory are different measures. Dead stock is zero sales across the company; broader unsold inventory captures stock remaining unsold under the notebook's inventory-flow definition.
- The inventory reconciliation result is unusually clean, so the notebook recommends retesting this logic on another real-world data source.
- Promotions are too rare in the source data to support a reliable effectiveness conclusion.